

Bruno Garcia

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EDUCATION

University of Michigan, Ann Arbor, MI May 2028
Bachelor of Science, Computer Science
Coursework: Data Structures and Algorithms, Object Oriented Programming, Discrete Math, Web Systems

EXPERIENCE

Cambridge Mobile Telematics June 2026 – August 2026
Software Engineer Intern Cambridge, MA

- Improved performance for the telematics analytics API by caching repeated table initialization for 6 hours, **~35% fewer database calls** and faster dashboard and chart queries across large customer datasets.
- Built and deployed **MCP tools** with **Terraform** that let an **AWS DevOps agent** query a time-series metrics database for anomalies, reducing root-cause analysis time by **~80%** for ride failures and device-update issues.
- Built a lightweight **Python polling system** for an analytics chart server that refreshed data every minute, served **1,000+ daily chart requests**, and reduced repeated backend requests substantially reducing computational costs.
- Built a simple rule based **anomaly-detection algorithm** in **Python** using **pandas** and **DuckDB** to scan **400,000+ driver records** across rolling 3-day windows and surface ride-failure patterns for further investigation.

Michigan Medicine May 2025 – Aug 2025
Software Engineer Intern Ann Arbor, MI

- Built an internal pharmaceutical analytics platform with **Python**, **FastAPI**, and **PostgreSQL**, allowing researchers to explore dispensing trends, and filter results without writing SQL or manually exporting data for analysis.
- Reduced dashboard load time by **60%** across **500K+ dispensing records** using **PostgreSQL** indexes, and **Redis** caching, cutting repeated database work and making medication and cohort views faster to use.
- Improved platform reliability with **FastAPI** validation, **PostgreSQL** and access controls, catching bad dispensing records, restricting study access, and reducing errors researchers encountered using the internal analytics platform.

PROJECTS

PomodoroLB | JavaScript, Node.js, Next.js, PostgreSQL, Supabase, Vercel

- Built a **Next.js** Pomodoro app where users could track study sessions and compare study time through weekly, monthly, and all-time leaderboards, reaching **20+ users** and **50+ hours** logged.
- Added user accounts and session tracking with **Supabase** and **PostgreSQL**, ensuring completed study sessions were recorded correctly and reflected in leaderboard rankings.

Mini Git | C++, CMake, OpenSSL, zlib, GoogleTest

- Built a Git-compatible version control system in **C++**, implementing content-addressed object storage, **SHA-1 hashing**, zlib compression, staging, commits, branching, and working-tree restoration.
- Engineered Git-style blob, tree, and commit objects with filesystem-backed references and **DAG traversal**, supporting init, add, commit, status, log, branch, checkout, and diff.

Fortnite Item Tracker | Golang, Gin, PostgreSQL, Docker, Discord API

- Built and deployed a Discord bot and web app that alerts users when tracked cosmetics become available in the Fortnite item shop.
- Engineered secure authentication using Discord OAuth2 and JSON Web Tokens for session management.

SKILLS

Languages: Python, HTML/CSS, SQL, C++, C#, R, JavaScript
Tools: Git, SQLite, Docker, BeautifulSoup, REST APIs, JSON, Jenkins, Terraform
Libraries: Pandas, scikit-learn
Interests: Rock Climbing, Ping Pong, NBA